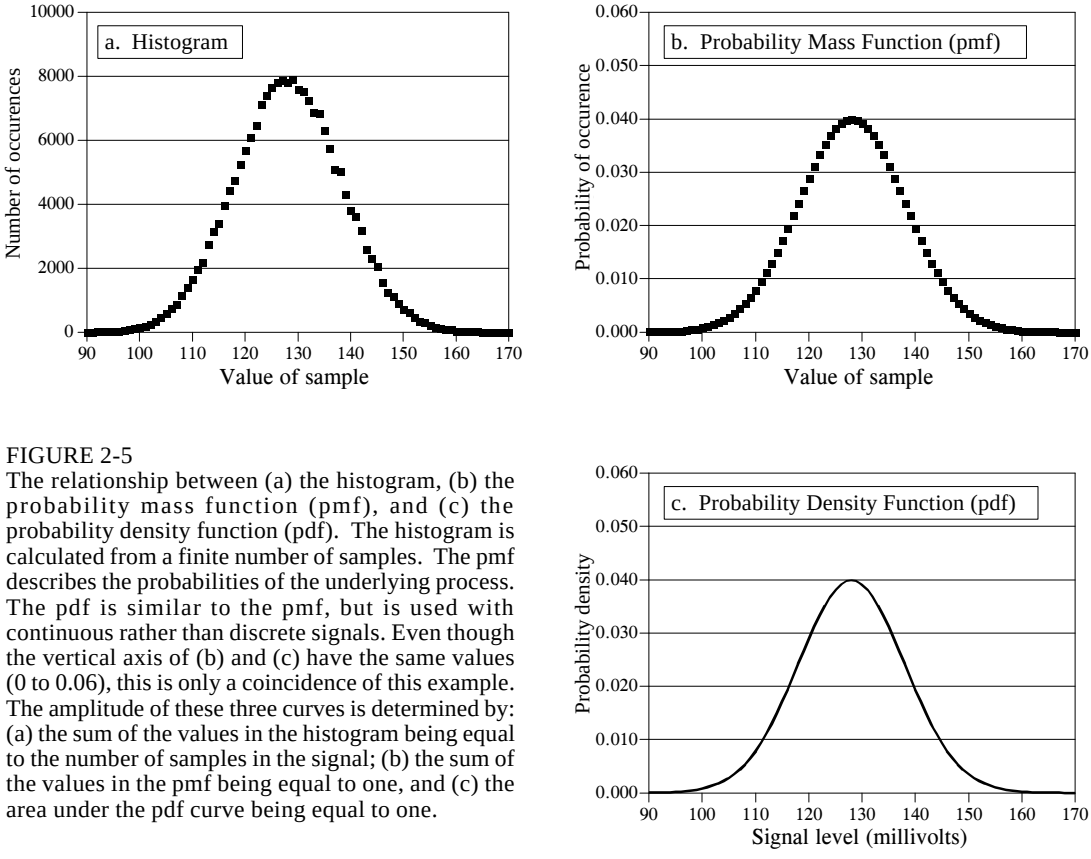

FIGURE 2-5
The relationship between (a) the histogram, (b) the probability mass function (pmf), and (c) the probability density function (pdf). The histogram is calculated from a finite number of samples. The pmf describes the probabilities of the underlying process. The pdf is similar to the pmf, but is used with continuous rather than discrete signals. Even though the vertical axis of (b) and (c) have the same values (0 to 0.06), this is only a coincidence of this example. The amplitude of these three curves is determined by: (a) the sum of the values in the histogram being equal to the number of samples in the signal; (b) the sum of the values in the pmf being equal to one, and (c) the area under the pdf curve being equal to one.

signal is shown by the *markers* in Fig. 2-5b. Similarly, the pdf of the analog signal is shown by the *continuous line* in (c), indicating the signal can take on a continuous range of values, such as the voltage in an electronic circuit.

The vertical axis of the pdf is in units of **probability density**, rather than just probability. For example, a pdf of 0.03 at 120.5 *does not* mean that the a voltage of 120.5 millivolts will occur 3% of the time. In fact, the probability of the continuous signal being exactly 120.5 millivolts is infinitesimally small. This is because there are an infinite number of possible values that the signal needs to divide its time between: 120.49997, 120.49998, 120.49999, etc. The chance that the signal happens to be exactly 120.50000··· is very remote indeed!

To calculate a *probability*, the *probability density* is multiplied by a *range* of values. For example, the probability that the signal, at any given instant, will be between the values of 120 and 121 is: $121-120 \times 0.03 = 0.03$. The probability that the signal will be between 120.4 and 120.5 is: $120.5-120.4 \times 0.03 = 0.003$, etc. If the pdf is not constant over the range of interest, the multiplication becomes the integral of the pdf over that range. In other words, the area under the pdf bounded by the specified values. Since the value of the signal must always be *something*, the total area under the pdf